# Bridge Construction Monitoring Based on TLS and Photogrammetry

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**Abstract.** Bridge construction monitoring is a key technology to control the quality and affect the life of a bridge. With the rise of Building Information Model (BIM), a series of new technologies have been applied to bridge construction monitoring. However, the practice of these new technologies is still in its early phase. At present, Terrestrial Laser Scanning (TLS) and photogrammetry are widely used monitoring methods. The former provides the high-precision spatial information about bridge components, while the latter provides the appearance information about them. The 3D simulation of the real state of the bridge can be obtained by the combination of the two. Based on TLS and photogrammetry, the construction monitoring technology of large bridge has been studied in this paper and applied in Taihong Bridge project. High-precision deformation monitoring, construction process control and surface crack detection are realized, which provides a reliable method for construction monitoring of large bridges.

## 1. Introduction

The construction process contains a huge production system, which needs multi-participation and multi-cooperation. Therefore, construction industry, compared to other non-farm industries, has been relatively slow in adopting new technologies and tools into its practices [1][2]. In surveying and mapping industry, terrestrial laser scanning (TLS) and photogrammetry emerge as the times require. The combination of TLS and photogrammetry, which we can call reality capturing technology, precisely realizes the reality simulation, thus stimulating the new cooperation system of building information model (BIM) [3][4]. With real simulation and dynamic tracking based on information model, BIM can transform the traditional cooperation mode which is mainly diachronic into synchronic, thus completing the innovation of production system integration.

In the aspect of bridge engineering, with the development of bridges in developed industrialized countries entering the ‘management and maintenance-oriented stage’ [5], the problem of bridge aging has been paid more and more attention. Therefore, the developing countries that bridge engineering is in the stage of ‘building and maintenance’ are facing higher requirements in the control of the construction quality of new bridges. At the same time, the breakthrough of large-scale bridge also needs the new construction monitoring technology to follow. Especially for long-span bridges, there are many working conditions in the construction process, the structure has strong geometric nonlinearity, and the means of error adjustment after the bridge completion are limited. Therefore, the accuracy of construction monitoring calculation and the provision of 3D visual monitoring information for decision makers are particularly important [6].

Based on BIM, this paper has studied the dynamic visualization of bridge construction control, combining with laser scanning, photogrammetry and 3D design model, forming a bridge construction monitoring information model which integrates all data. Further, the construction quality is visualized in the computer, so that all participants can understand all aspects of the whole bridge even if they are not on the site, and at the same time, the problems in the construction process can be controlled and handled in time, so as to achieve the real-time tracking of the whole bridge construction process and construction quality.

## 2. The technical routes

Before the work is officially carried out, it is necessary to study and summarize *the overall technical scheme for construction monitoring of Taihong bridge*, so as to form a forward technical route and a reverse technical route. The combination of forward and reverse can complete the dynamic correction and task linkage of the whole construction monitoring process.

### *2.1. The forward technical route*

In this project, construction monitoring is divided into six stages (Figure 1).

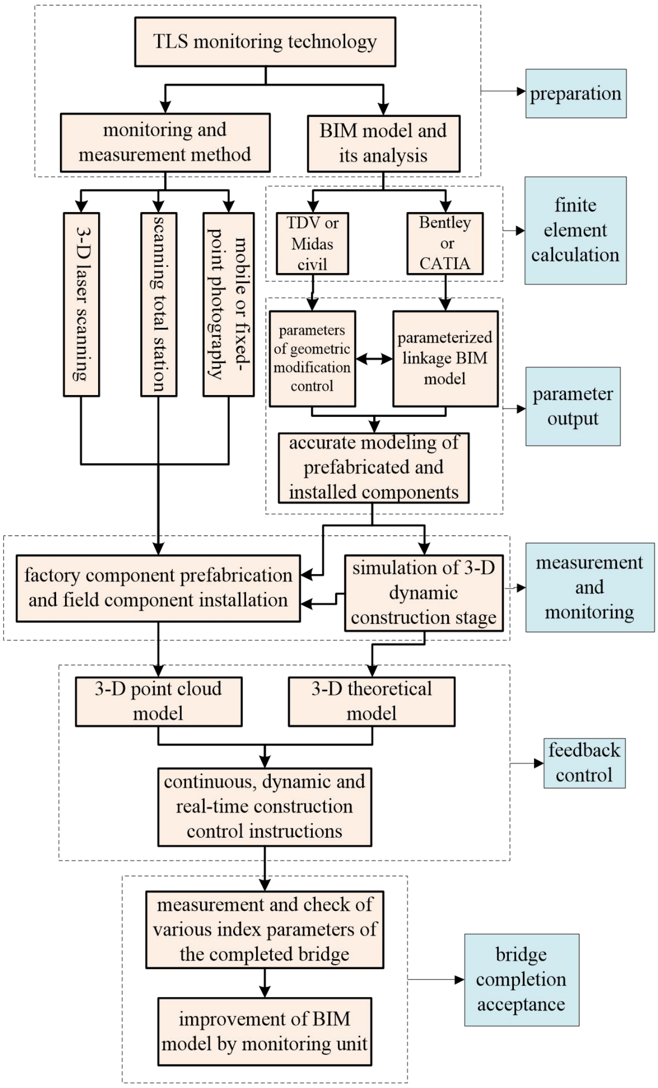


**Figure 1.** The forward technology route based on BIM

*2.1.1. Preparation.* Led by the owner, the design and construction unit are responsible for building BIM model of bridge design with accurate parameters and accurate BIM model of construction stage simulation. The construction monitoring unit shall review the initial designed BIM model, be familiar with the BIM model of construction simulation, and prepare the construction monitoring plan and implementation rules. At the same time, according to the characteristics and construction methods of the bridge, TLS measuring equipment is selected to adjust the measurement accuracy.

*2.1.2. Finite element calculation.* The calculation accuracy of the finite element model directly affects the accuracy of the completed bridge. In the traditional construction monitoring process, the finite element calculation parameters used for monitoring come from the two-dimensional design drawings, and the construction process comes from the paper construction organization design data provided by the construction unit. In the process of finite element modelling, some parameters are ambiguous, or the values of design and construction are contradictory, or the parameters are missing, which makes the calculation results of finite element difficult to be consistent with the expected results of design[7][8].

The BIM model based on unified data source avoids the above problems. The monitoring unit directly extracts the design structural parameters from the database of BIM model, and extracts the material mechanical parameters corrected by the construction unit. At the same time, according to the construction stage simulation formulated by the design and construction unit, the finite element simulation of the construction stage can be carried out accurately. Furthermore, according to BIM spatial geometry model and statistical function, accurate weight statistics and finite element component volume weight correction can be carried out for each component of long-span bridge, which will directly avoid the calculation error caused by the statistical error of self-weight of long-span bridge.

*2.1.3. Parameter output.* After the finite element analysis from the unified data source is completed, the theoretical calculation results will be obtained for each construction stage and the finished bridge state, mainly including:

- unstressed dimensions of components;
- internal force, stress and displacement of each construction stage;
- internal force, stress and displacement of the finished bridge state.

These theoretical calculation results will be updated in BIM model. According to the unstressed dimensions, the geometric models of components in BIM will be updated precisely to provide theoretical model for intelligent manufacturing. According to the theoretical displacement of each construction stage, the geometric parameters indicating the bridge displacement are extracted and updated to the BIM construction stage model. The theoretical stress and strain of each member in each construction stage should be updated to the database of BIM model to facilitate real-time retrieval.

*2.1.4. Measurement and monitoring.* When the bridge starts to be constructed, the measuring group responsible for monitoring shall be brought into the site. According to the characteristics of TLS method, suitable measuring stations are selected and the measuring coordinate system of the bridge is established. After the introduction of 3D laser scanning technology, the scanning mode is determined according to the accuracy of the instrument and the required accuracy of the bridge. Generally, the accuracy of fixed-point scanning will be more assured [9][10]. Then, an automatic data acquisition and analysis system is established to acquire the point cloud model, which is directly compared and analysed with the current theoretical model on BIM platform, so as to obtain construction control errors.

*2.1.5. Feedback control.* By using BIM platform and TLS to obtain the 3D geometric state of the bridge, the deviation between the actual and theoretical state of the bridge can be derived, including local deviation, overall deviation, axial deviation, transverse deviation, corner deviation, etc. All deviations will be carefully analysed and evaluated in the 3D state, and the construction monitoring instructions in subsequent stages will be corrected by feedback to ensure that the bridge is strictly close to the theoretical state in the 3D state. Based on BIM platform, closed-loop construction monitoring of large bridges can be accurately achieved [11].

*2.1.6. Bridge completion acceptance.* In the whole life cycle of large bridges, the bridge completion acceptance stage is the only stage that can accurately and comprehensively obtain the status information of the structure. The construction monitoring controls the change of the state information from construction to completion. Through continuous analysis, simulation, measurement and feedback of the whole process of construction monitoring, the information of the finished bridge state can accurately reflect the reality, especially the mechanical state of the bridge. Based on BIM construction monitoring, all data of construction stage and completion of the bridge will be updated to BIM model, which will be the data basis for safe operation, health monitoring, maintenance and reinforcement of the bridge [12].

### *2.2. The reverse technical route*

As a dynamic tracking technology, the TLS workflow is not one-way, but cyclical. Therefore, combined with bridge data and BIM model, the technical scheme of reverse impact is studied. This route is divided into component prefabrication and construction assembly.

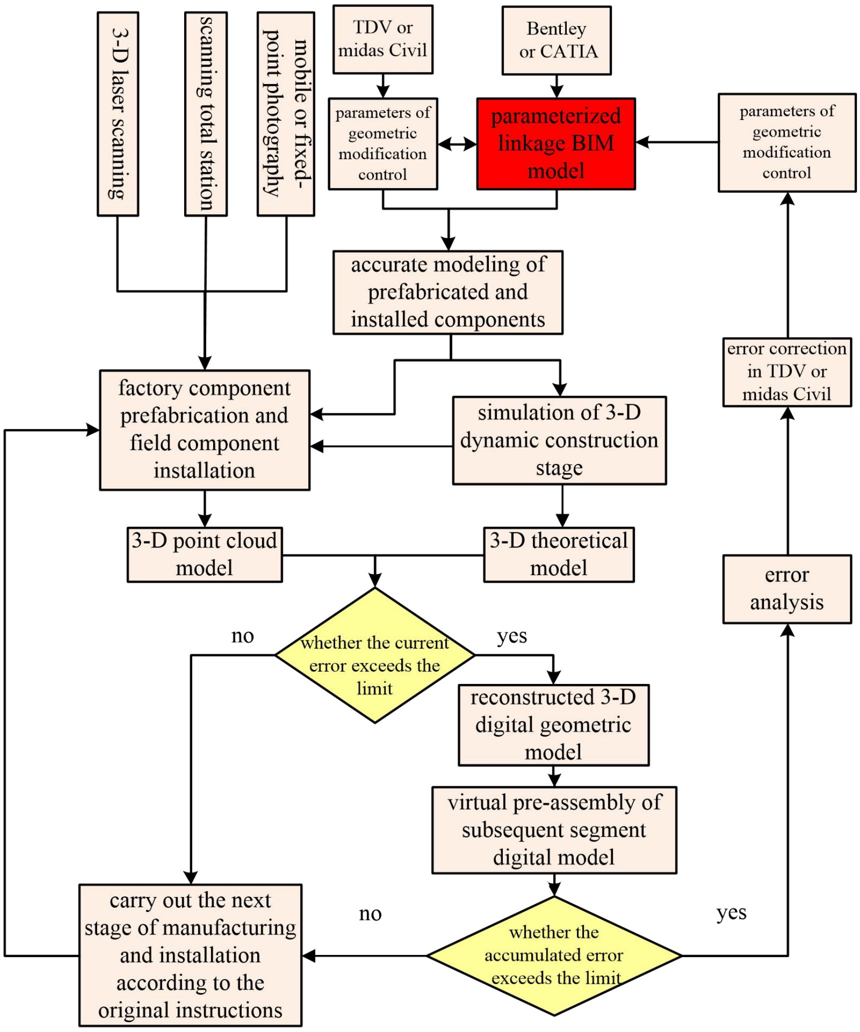


**Figure 2.** The reverse technical route of construction monitoring based on BIM

*2.2.1. component prefabrication.* In this stage, according to TLS and 3D reconstruction technology, virtual pre-assembly of prefabricated components can be realized to judge whether the prefabricated components meet the design requirements. If the error exceeds the limit, the manufacturing dimension of the subsequent prefabricated components will be adjusted according to the results of virtual pre-assembly to ensure that the error during bridge completion is within the controllable range.

*2.2.2. construction assembly.* In this stage, according to TLS and 3D reconstruction technology, the digital model which is consistent with the current stage of the bridge can be obtained directly, and the subsequent segment prefabricated component digital model obtained by 3D reconstruction can be pre-assembled virtually to further determine whether the subsequent segment meets the theoretical error requirements and give the subsequent segment adjustment instructions.

## 3. Theoretical research and practical application

*3.1. Accuracy improvement of laser scanning and point cloud splicing*

By mastering the data type, acquisition efficiency and result accuracy of current mainstream scanners, a scanning acquisition scheme meeting the efficiency and accuracy is put forward for different extreme situations in engineering practice. The scanners studied include mainstream instruments such as Faro 330, Faro 350, Leica MS60 and Leica P50. See Figure 3 for an illustration. The application scenarios studied include high-altitude scanning, remote spacer scanning, large-volume structure scanning, special-shaped structure scanning, large-scene component scanning, large-component panel scanning, stable scanning in vibration environment, rainy day scanning, high-temperature and strong light scanning, and night scanning. In the scanning environment above, high accuracy and high efficiency acquisition of 3D scanning point cloud data can be achieved through fine scanning scheme design [13].


**Figure 3.** Laser scanners

At this stage, target-free high-precision splicing method was innovatively proposed to solve the problems of accurate splicing of large-scale complex structure multi-site cloud and construction coordinate system. In complex environment, traditional spherical target splicing method makes the number of scanning stations and the scale of scanning data increase dramatically, and the comprehensive splicing accuracy of large and complex structures achieve only cm-level due to on-site shielding and comprehensive influence of target breakage caused by construction uncertainties [14]. Traditional direct splicing method based on point cloud for target-free has strict requirements on overlap of scanning point cloud, initial position of point cloud and shape of point cloud [15]. We've proposed that using Leica MS60 for uniform construction coordinate system scanning and Faro 350 for main structure scanning. Then, point cloud data of all kinds of instruments is spliced accurately by feature extraction and attitude adjustment of point cloud data, which realizes high-precision point cloud data collection and splicing with construction coordinate system.

*3.2. High-precision deformation monitoring*

In order to realize accurate monitoring of spatial deformation of large-span and complex structures, an accurate monitoring method based on high-speed 3D scanning technology is proposed. By analysing the influencing factors of measurement accuracy when using 3D scanning, the relation of deformation measurement error is derived under the condition that the scanning station remains unchanged. The analysis shows that when the distance between scanning points meets the requirements, the deformation error will be less than 0.1mm. An experiment on measuring precision of deformation scanning of structure is designed (Figure 4).

**Figure 4.** Experiment on the accuracy of deformation scanning measurement

The experimental results show that the 3D scanning has the deformation monitoring accuracy of 0.1mm and the scanning accuracy is not affected by the scanning distance within the scanning range of 100m. Practice also shows that this method has the ability to accurately monitor 3D continuous deformation of bridge structure, and the monitoring accuracy reaches 0.3mm, thus realizing the goal of 3D real-time monitoring and evaluating safety of large-span structure. The relation between density of point cloud and accuracy of deformation monitoring is shown in Figure 5.

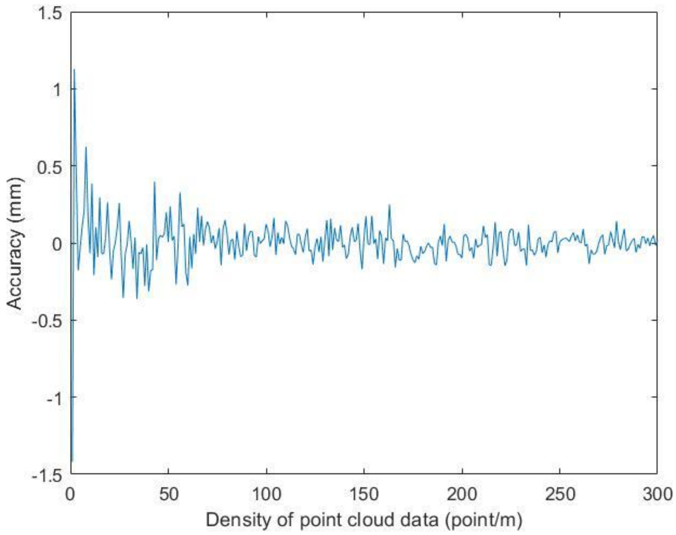


**Figure 5.** The relation between density of point cloud and accuracy of monitoring

*3.3. 3D model reconstruction and modification*

BIM model has the advantages of comprehensive information and small data volume. It is a unified data model for the industry in the future. However, the design of BIM model is inconsistent with the actual construction of physical structure in terms of space, size and project, which makes it difficult for BIM model to really guide the operation and maintenance of the structure. The scanning point cloud is used to collect the complex structure accurately, and the reverse reconstruction of BIM model based on scanning point cloud and the fast modification of existing BIM model are studied, which improves the application scope of BIM model. Figure 6 shows the reverse reconstruction of BIM model based on scanning point cloud.

**Figure 6.** Reverse reconstruction of design model using point cloud data

*3.4. Construction of 3D real-scene model of bridge*

Point cloud data can accurately reflect the geometrical and spatial dimensions of the structure, while high-definition images contain visual texture information and reflect the damage and defects of the structure. The research uses camera to collect high-definition images of bridge construction process, and establishes 3D real-scene model theory of bridge structure based on disordered images [16]. A 3D realistic model of the bridge has been built based on a large number of disordered images. Figure 6 shows a 3D realistic model based on high-definition images.

**Figure 6.** Experiment on the accuracy of deformation scanning measurement

*3.5. Crack identification and location*

Based on the 3D realistic model, the apparent damage information of the structure can be tracked accurately, and the high-precision measurement and positioning can be realized by algorithm. Figure 7 is a damage identification and location map of a realistic model.

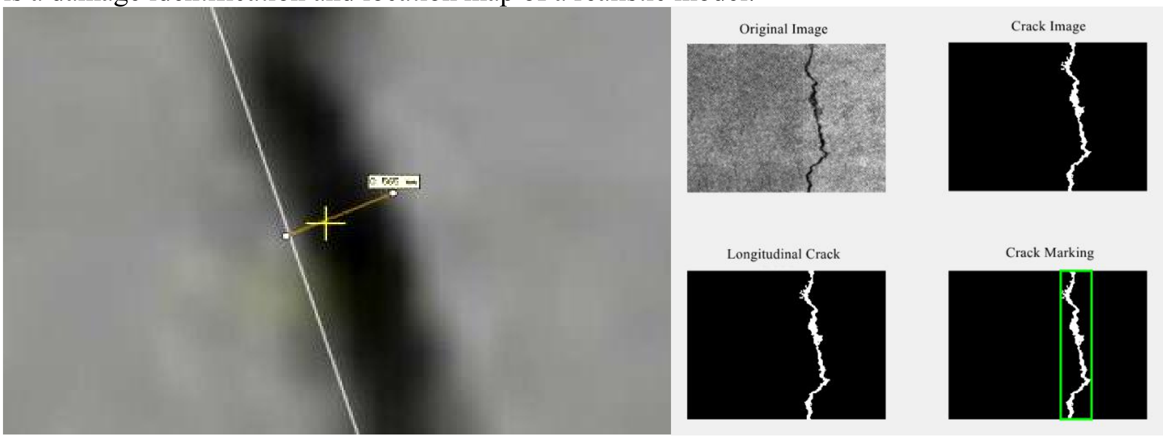


**Figure 7.** Fracture identification and location of real-scene model

## 4. Conclusion

The construction monitoring of long-span bridges is a necessary technical method to ensure the safety and reliability of the bridge from the digital model to the actual bridge. It is not only to ensure the safety of the bridge in the process of construction, but also has far-reaching significance:

- It will ensure that all important indexes after the completion of the bridge meet the expected requirements of the design;
- Ensure that the internal force state of the completed bridge meets the reasonable completion state of the design;
- In essence, the operation safety and service life of the bridge are improved.

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